# Lec-16

01/03/26 4:40 PM

0 1 1 2 3 5 8 13 21 34 55 - - -

Nth

$a^0 \to 1$ $(a^{n-1}) \times a$

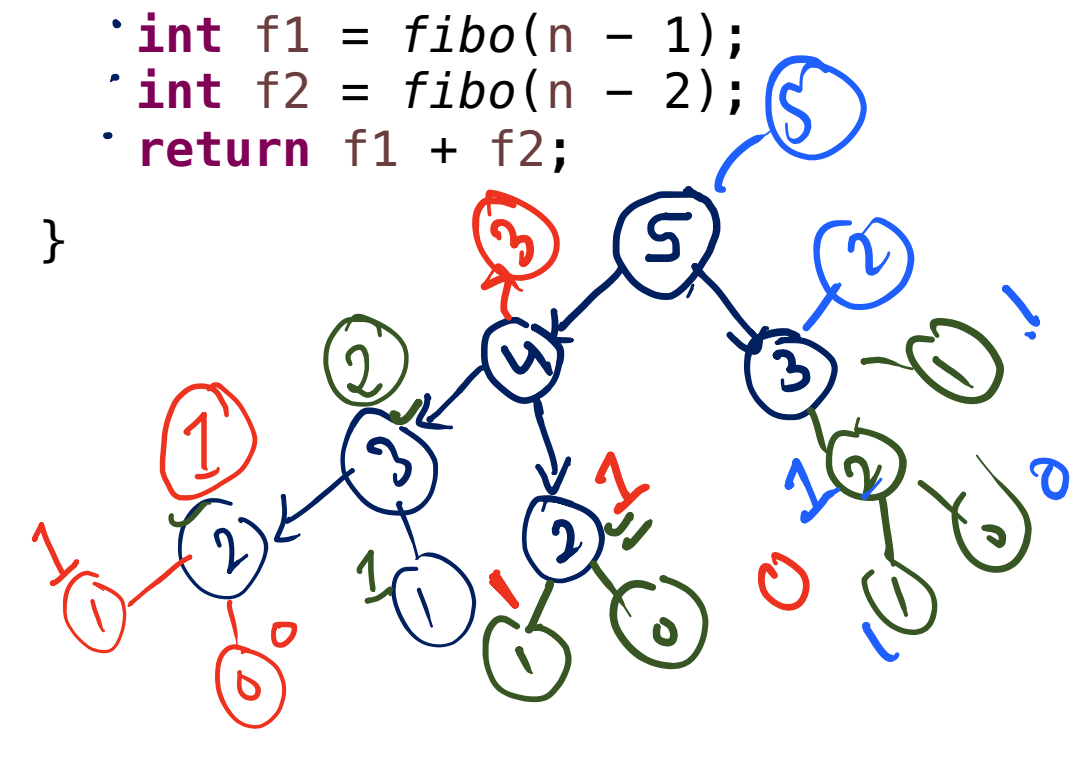
$0! = 1$ $(n-1)! \times n$

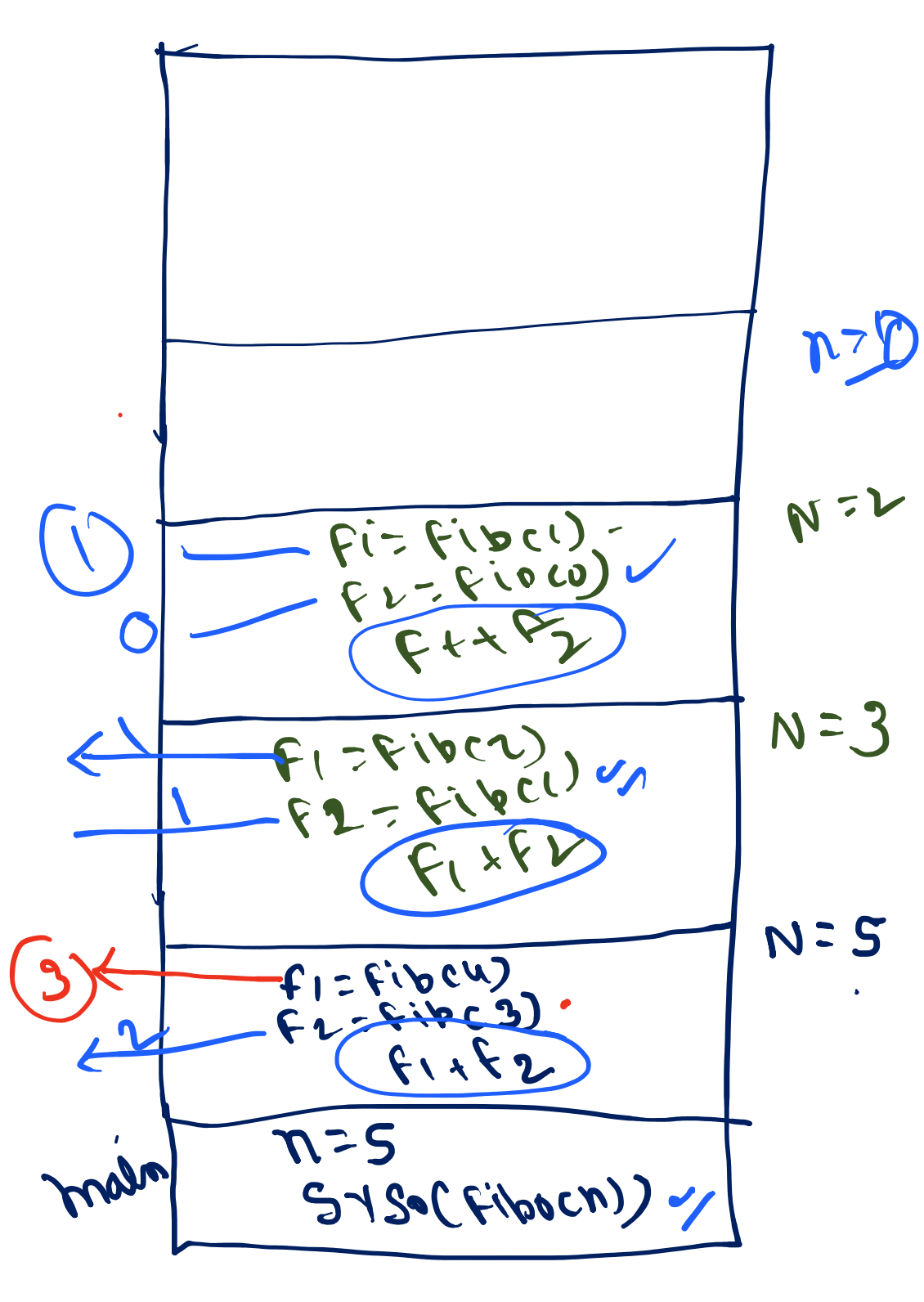
$n=0 \to 1$ $(n-1)! n$

$F_n = F_{n-1} + F_{n-2}$ sum

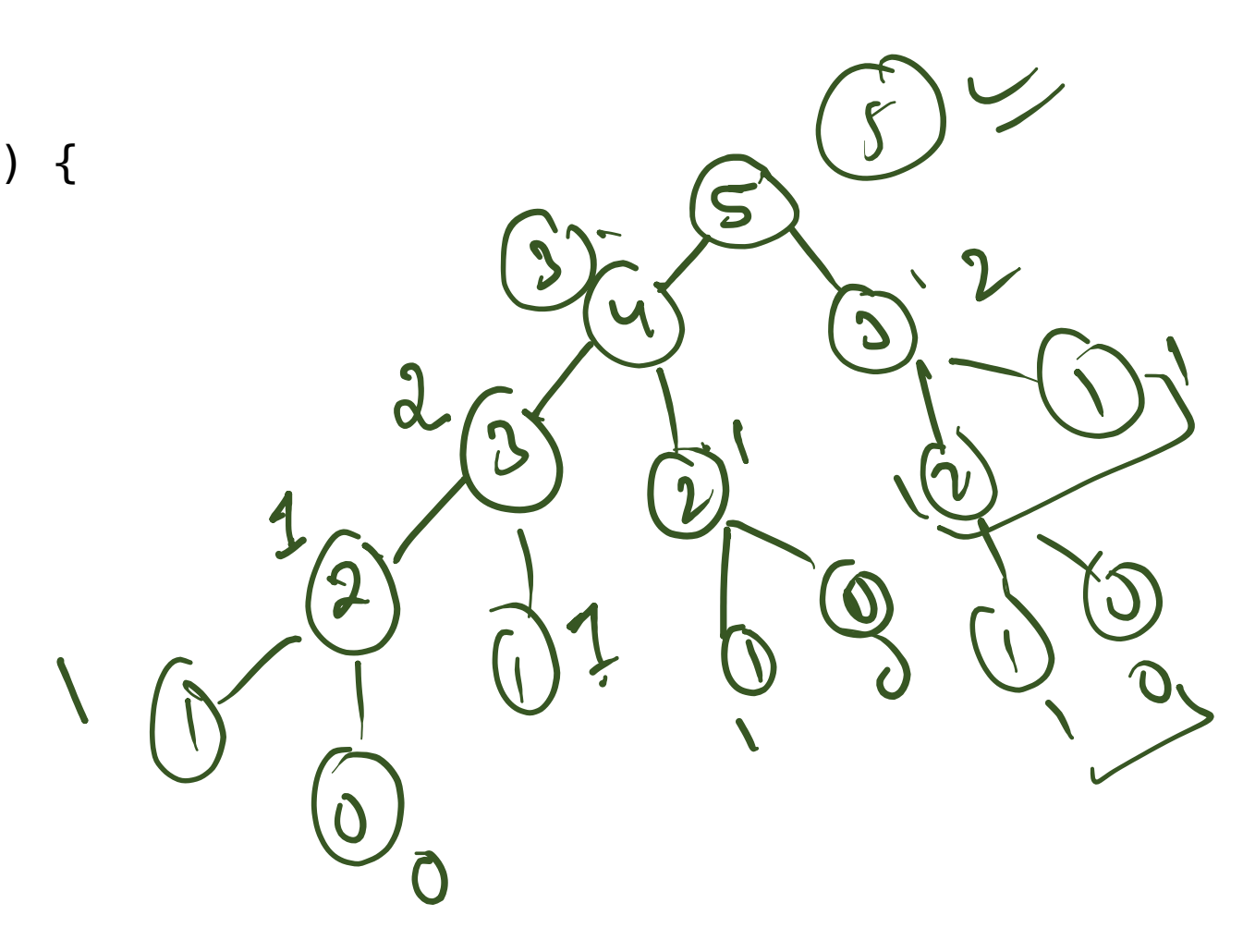
$N=2$ $f_1 + f_0$

```
public static void main(String[] args) {
        int n = 5;
        System.out.println(fibo(n));
    }

    public static int fibo(int n) {
        if (n == 0 || n == 1) {
            return n;
        }
        int f1 = fibo(n - 1);
        int f2 = fibo(n - 2);
        return f1 + f2;
    }
```


5
5
3
4
3
2
2
1
2
1
1
2
1
1
1
0
0
1
0
1
1
0
0


n=0
N=2
N=3
N=5
1
0
f1=fibo(1)
f2=fibo(0)
f1+f2
1
f1=fibo(2)
f2=fibo(1)
f1+f2
3
2
f1=fibo(4)
f2=fibo(3)
f1+f2
main
n=5
Syso(fibo(n))

```
public static void main(String[] args) {
        int n = 5;
        System.out.println(fibo(n));
    }

    public static int fibo(int n) {
        if (n == 0 || n == 1) {
            return n;
        }
        int f1 = fibo(n - 1);
        int f2 = fibo(n - 2);
        return f1 + f2;
    }
```


5
5
3
4
3
2
2
1
1
2
1
1
1
1
1
0
0
1
0
0
1
0
1
0

all subsequences

subsequence

String → abc

Subsequence ←

Substring ←

al pcd

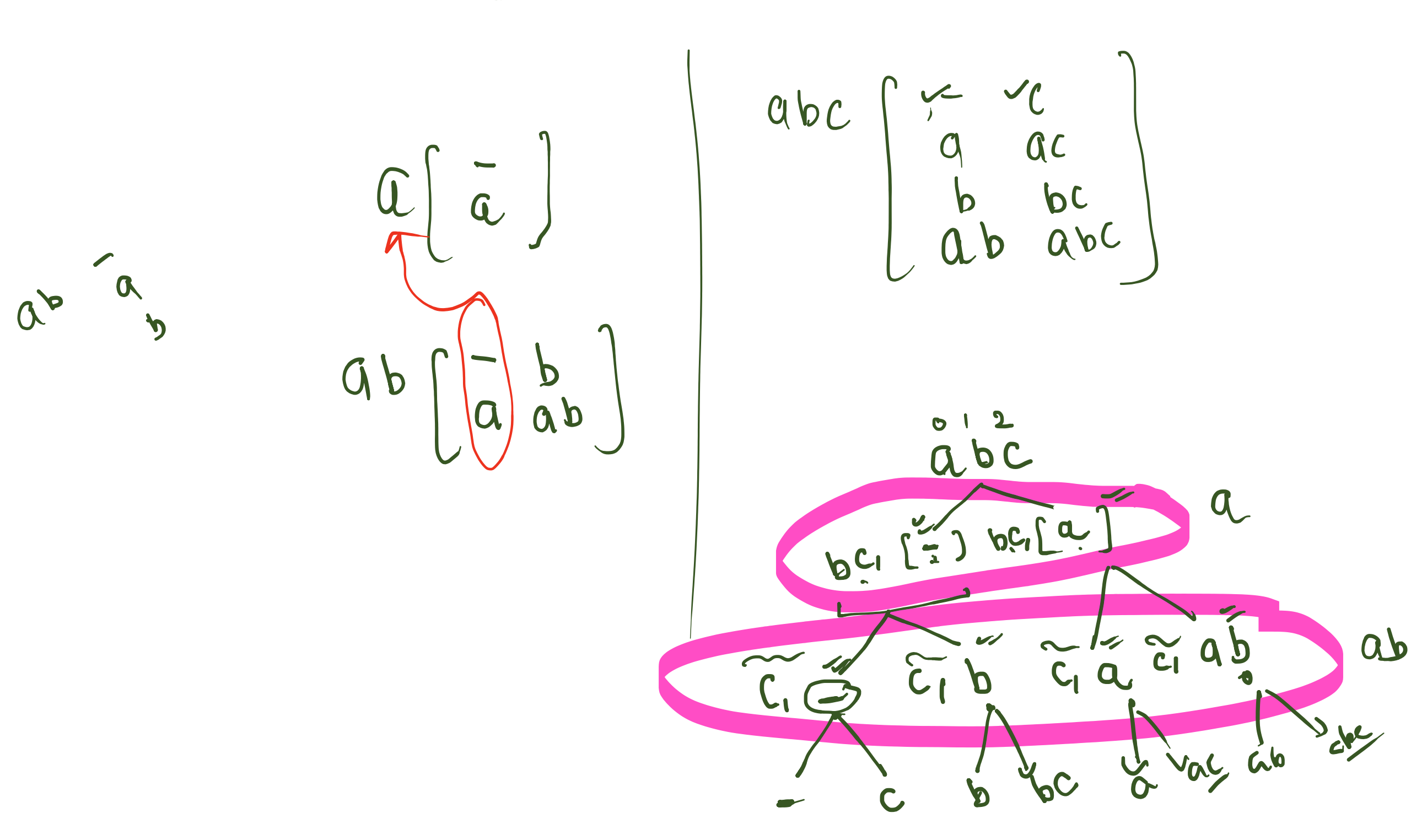
a pcd

Continue

alp

3

abc ⇒ -, a, b, c, ab, bc, ac, abc → 8 $2^3$


ab - a b
a [ - a ]
ab [ - a b ab ]
abc [ - a b ab c ac bc abc ]
0 1 2
abc
bc, [-] bc, [a]
a
c, - c, b c, a c, ab
ab
- c b bc a ac ab abc

[abc] → a, b, c, ab, bc, ac, abc $2^3 = 8$

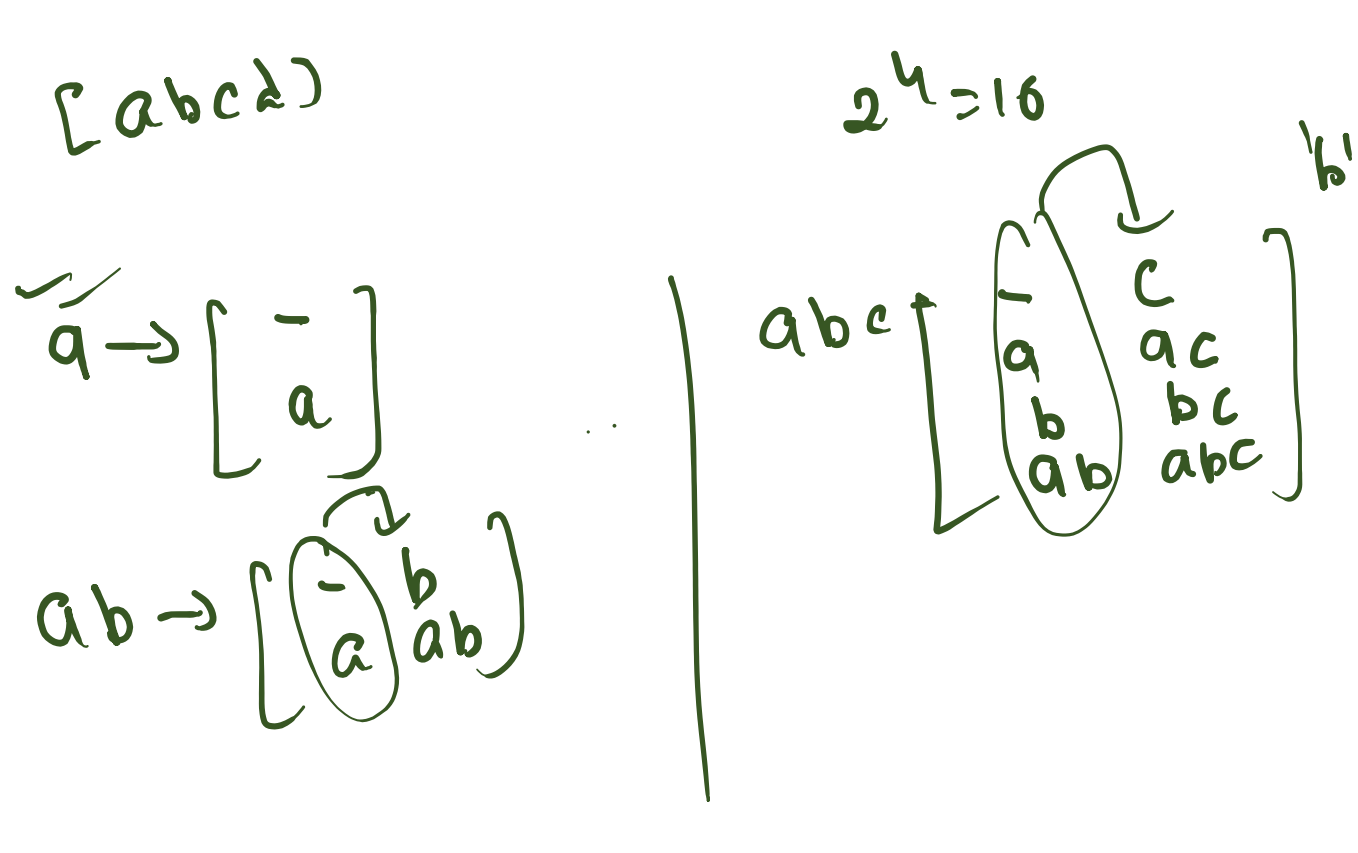
Size os = ""

ans(ch)

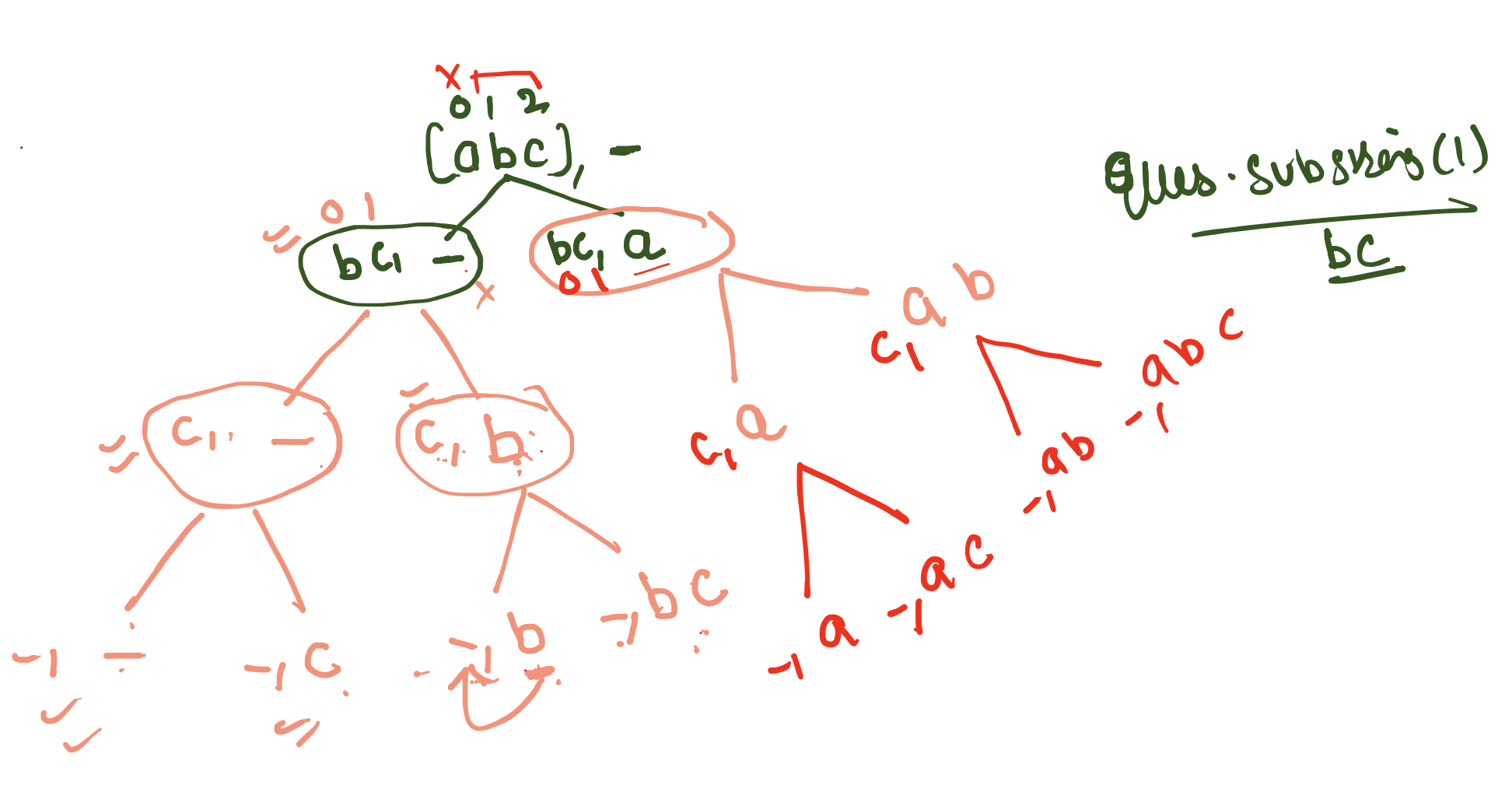
-+'b' = b

'b'+'c' = (bc)

[abcd] $2^4 = 16$

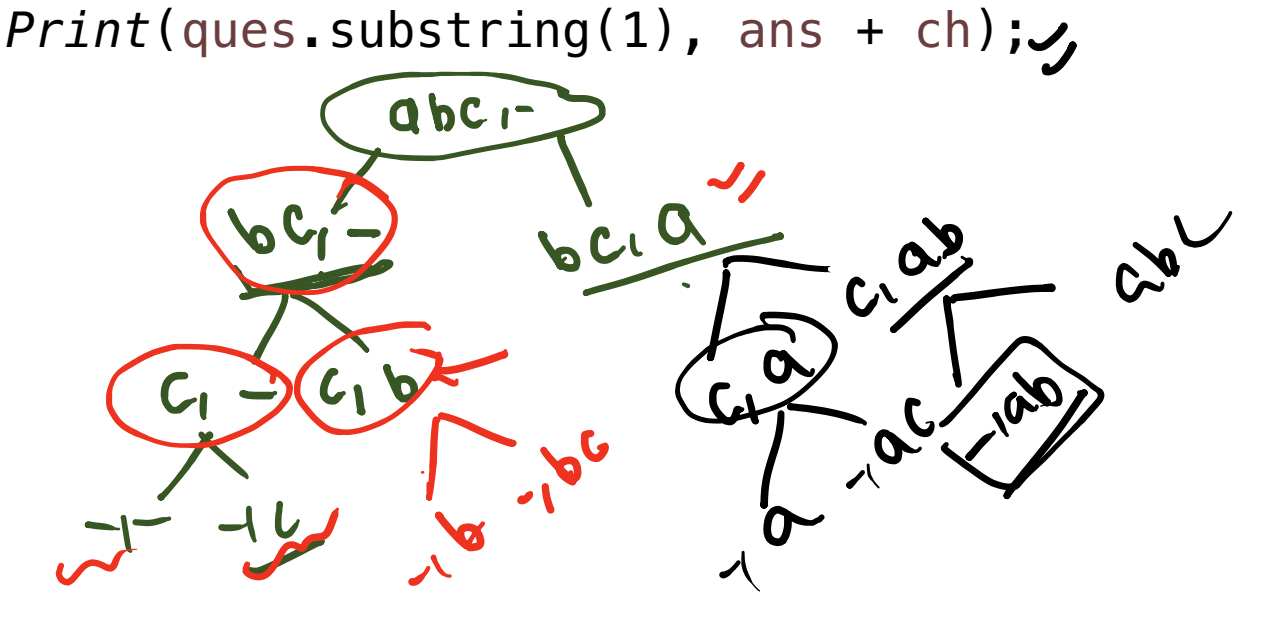

a → [ - a ]
ab → [ - a b ab ]
abc [ - a b ab c ac bc abc ]

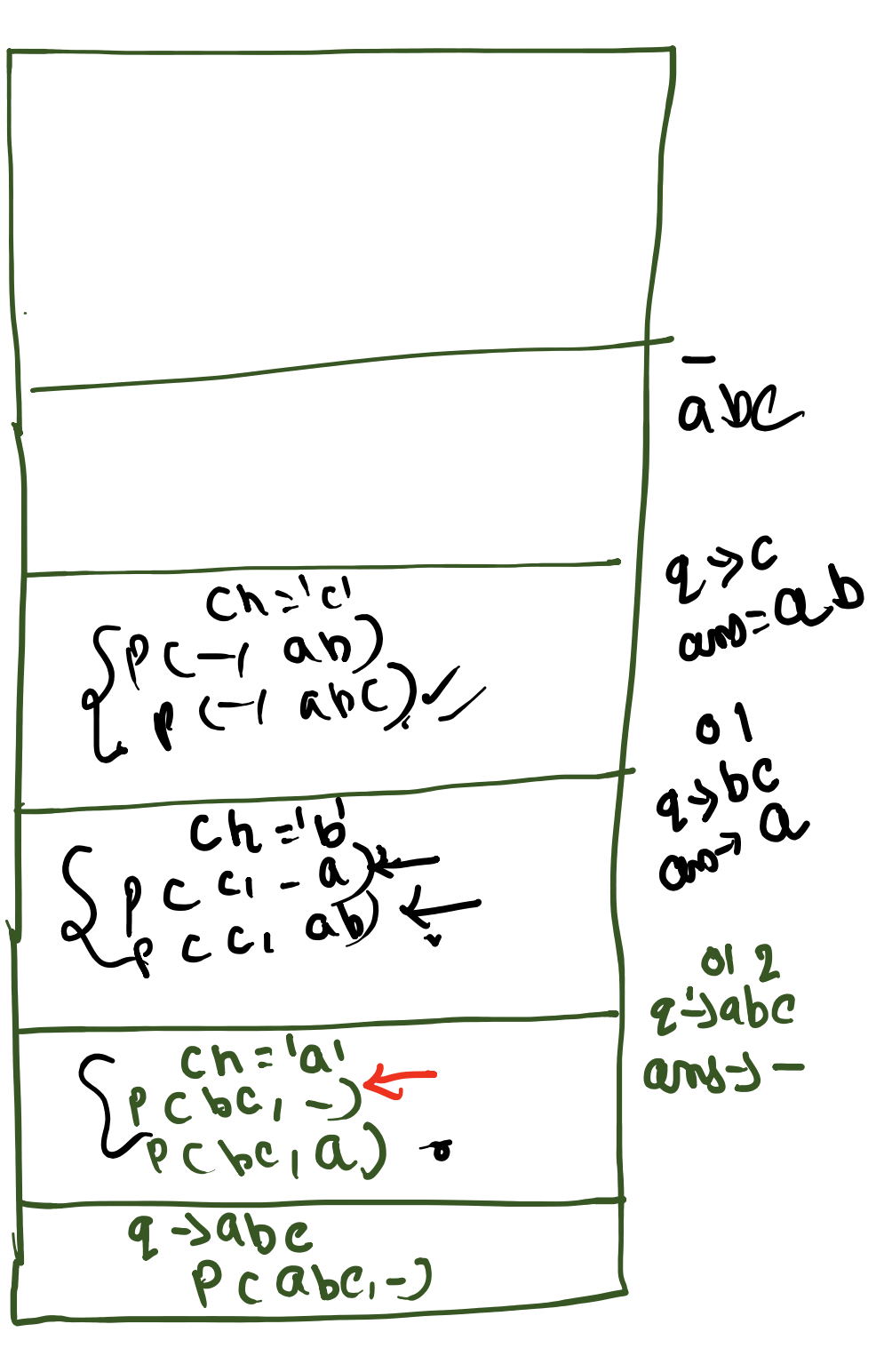
-
c
b
bc
a
ac
ab
abc

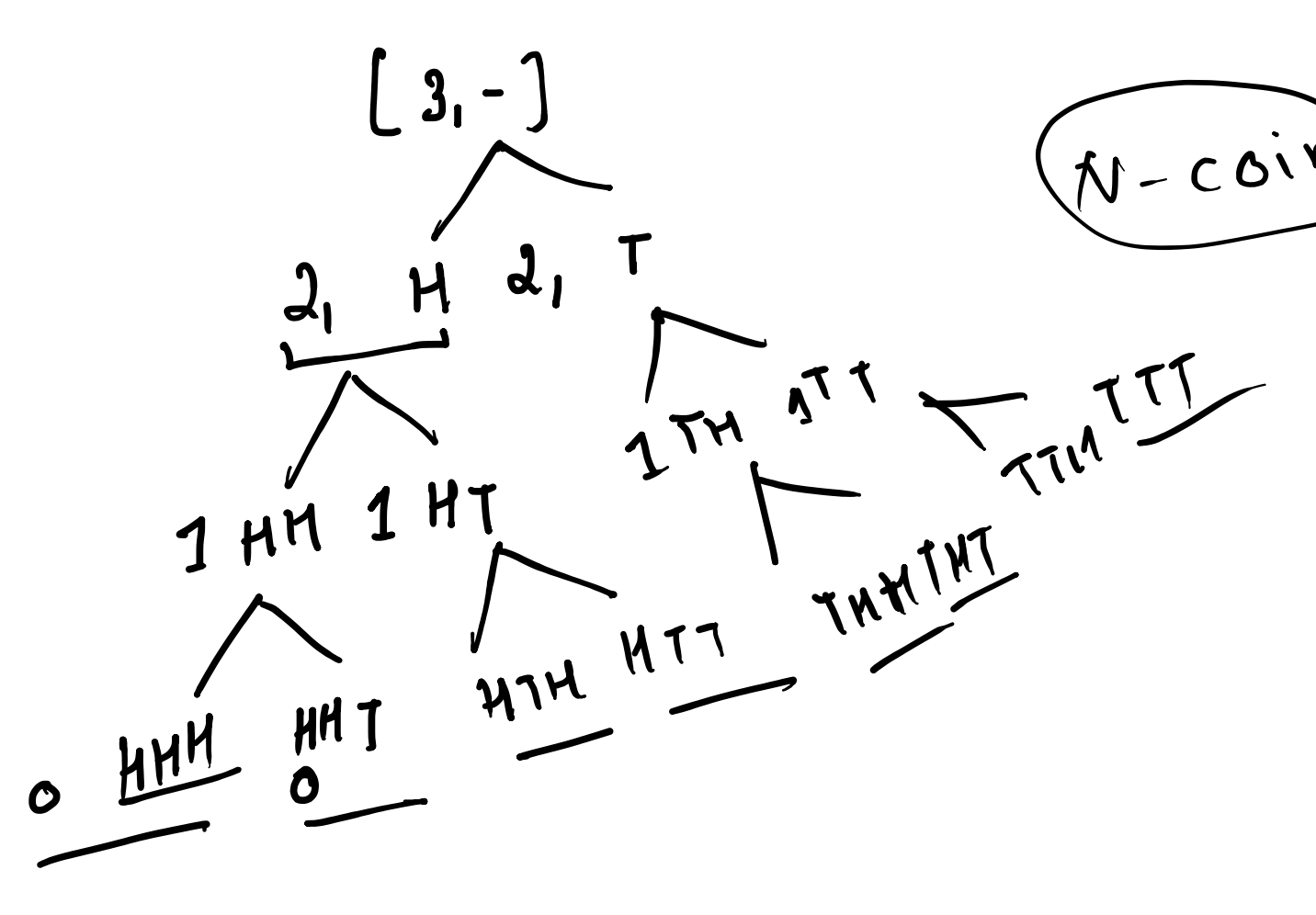

0 1 2
[abc], -
0 1
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
-, a
-, ac
-, ab
-, abc
Ques. substring(1)
bc

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String ques = "abc";
    Print(ques, "");
}

public static void Print(String ques, String ans) {
    if (ques.length() == 0) {
        System.out.println(ans);
        return;
    }
    char ch = ques.charAt(0);// 'a'
    Print(ques.substring(1), ans);
    Print(ques.substring(1), ans + ch);
}
```

-, c, b, bc, a, ac, ab, abc


abc, -
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
-, a
-, ac
-, ab
abc

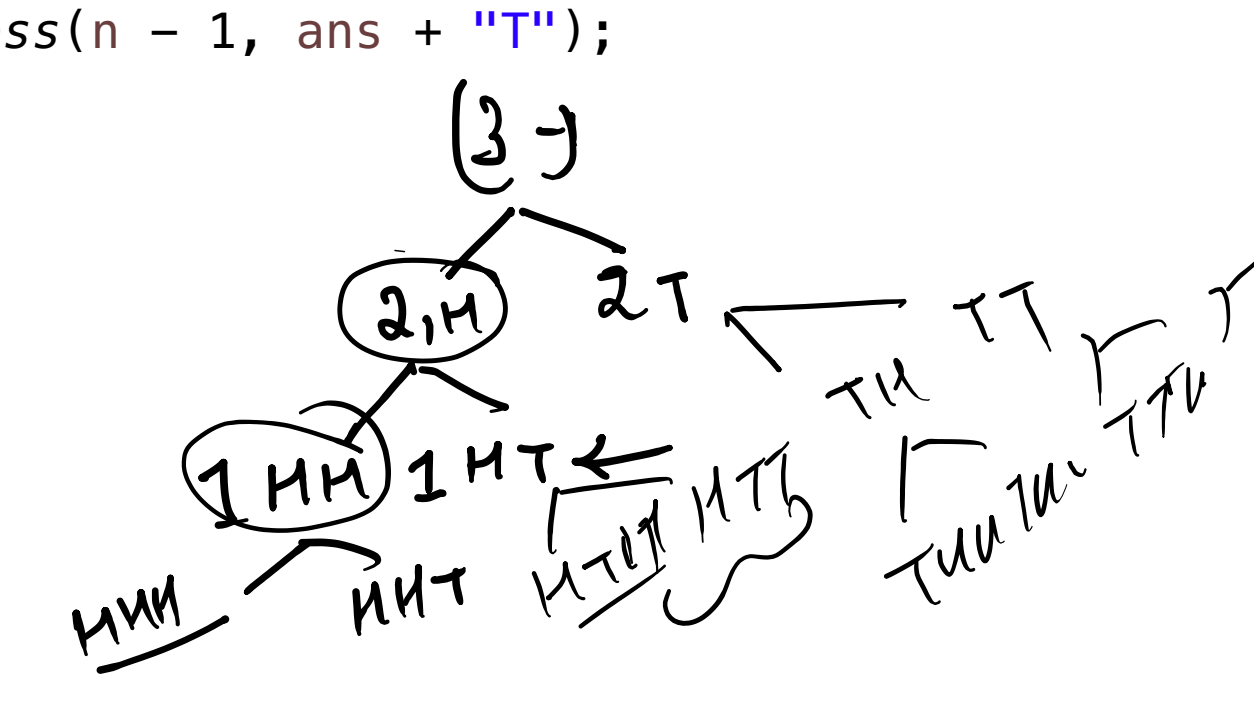

abc
a→c
ans=ab
0 1
a→bc
ans→a
0 1 2
a→abc
ans→-
ch='c'
P(-, ab)
P(-, abc)
ch='b'
P(c, -a)
P(c, ab)
ch='a'
P(bc, -)
P(bc, a)
q→abc
P(abc, -)

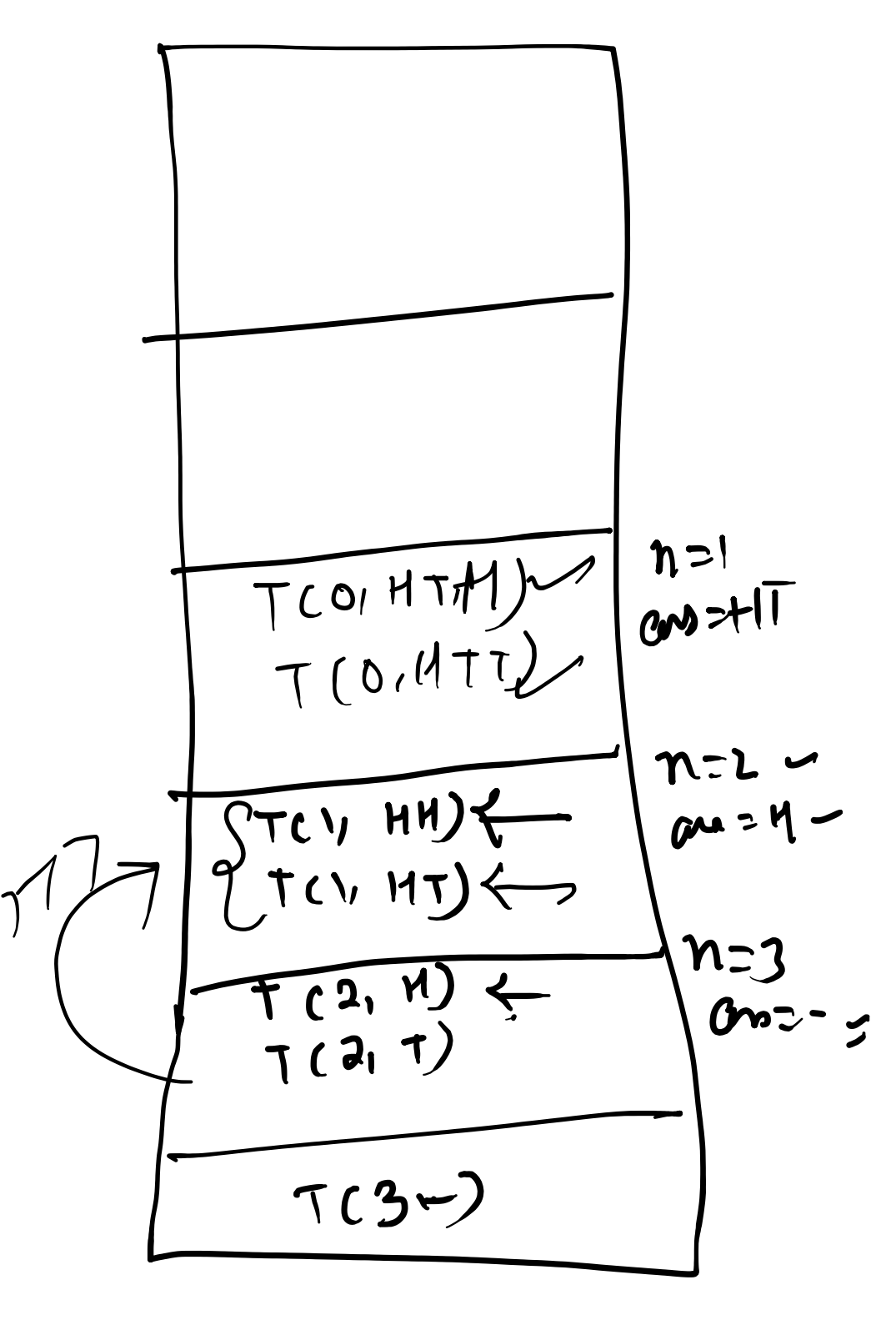

[3, -]
2, H
2, T
1 HH
1 HT
1 TH
1 TT
TTH
TTT
0 HHH
HHT
0
HTH
HTT
THH
THT

N-coin

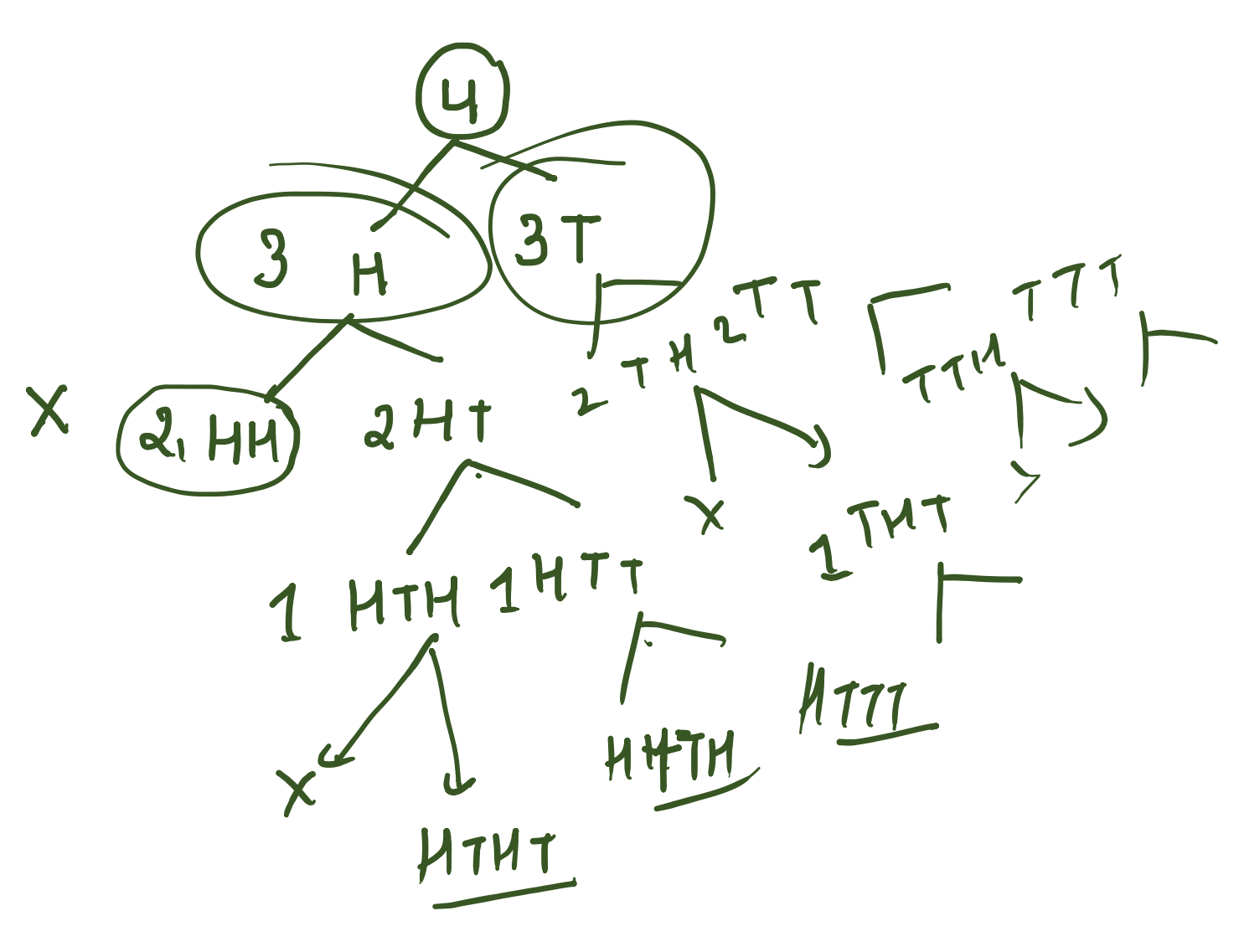
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 3;
    Toss(n, "");
}

public static void Toss(int n, String ans) {
    // TODO Auto-generated method stub
    if (n == 0) {
        System.out.println(ans);
        return;
    }
    Toss(n - 1, ans + "H");
    Toss(n - 1, ans + "T");
}
```


(3 -)
2, H
2 T
1 HH
1 HT
HHH
HHT
HTH
HTT
TH
TT
THH
THT
TTH
TTT


n=1
ans=HT
T(0, HTH)
T(0, HTT)
n=2
ans=H
T(1, HH)
T(1, HT)
n=3
ans=-
T(2, H)
T(2, T)
T(3-)

HHHH
HHHT
HHTH
HHTT
HTHH
HTHT
HTTH
HTTT
THHH
THHT
THTH
THTT
TTHH
TTHT
TTTH
TTTT

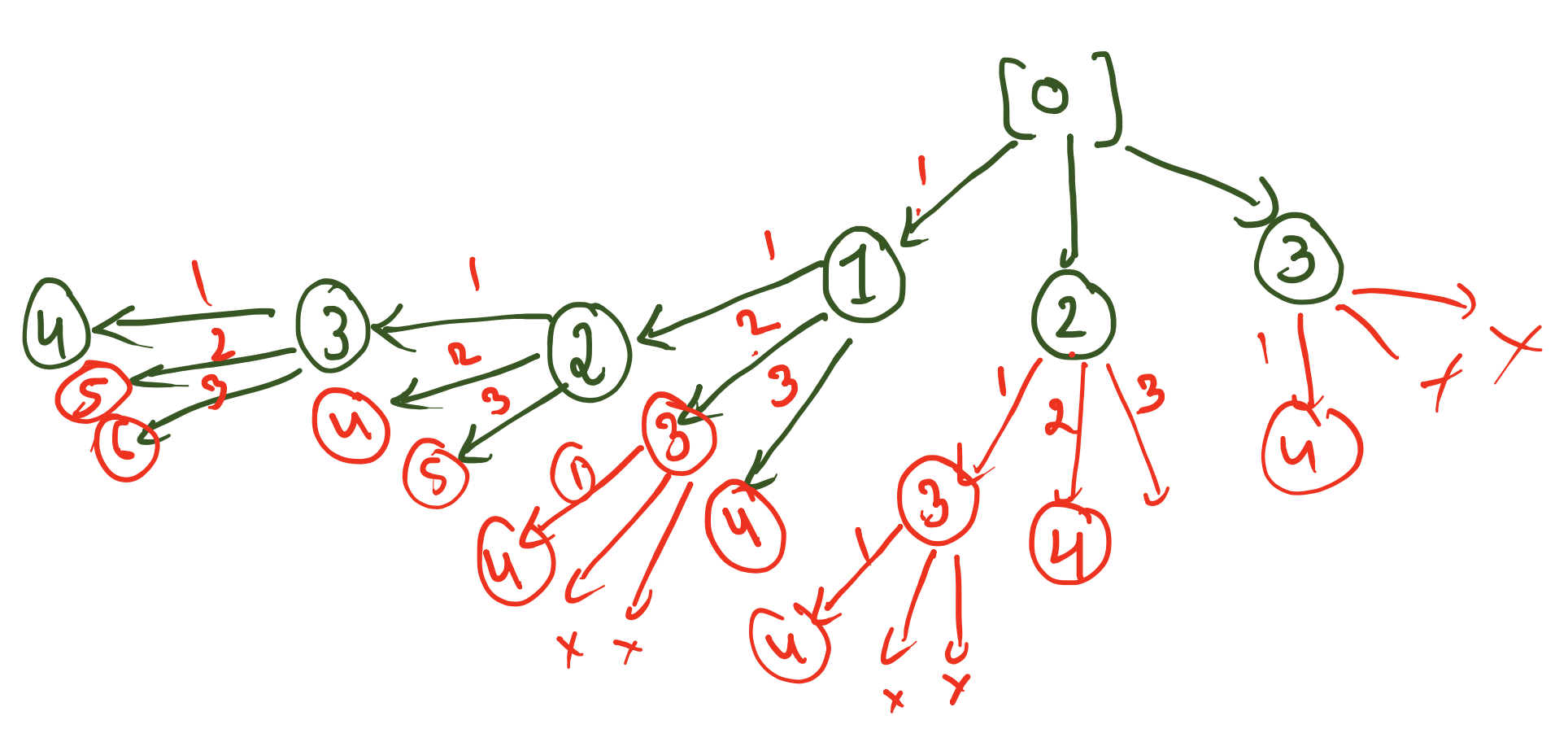

4
3 H
3T
X
2, HH
2 HT
2 TH
2 TT
TTH
TTT
1 HTH
1 HTT
1 THT
X
X
HTHT
HHTH
HTTT

1 2 3 4 5 6

dice = 3 → {1, 2, 3}

N=4

curr>N


0 1 2 3 4
[0]
1
2
3
1
2
3
4
5
6
X X
X Y

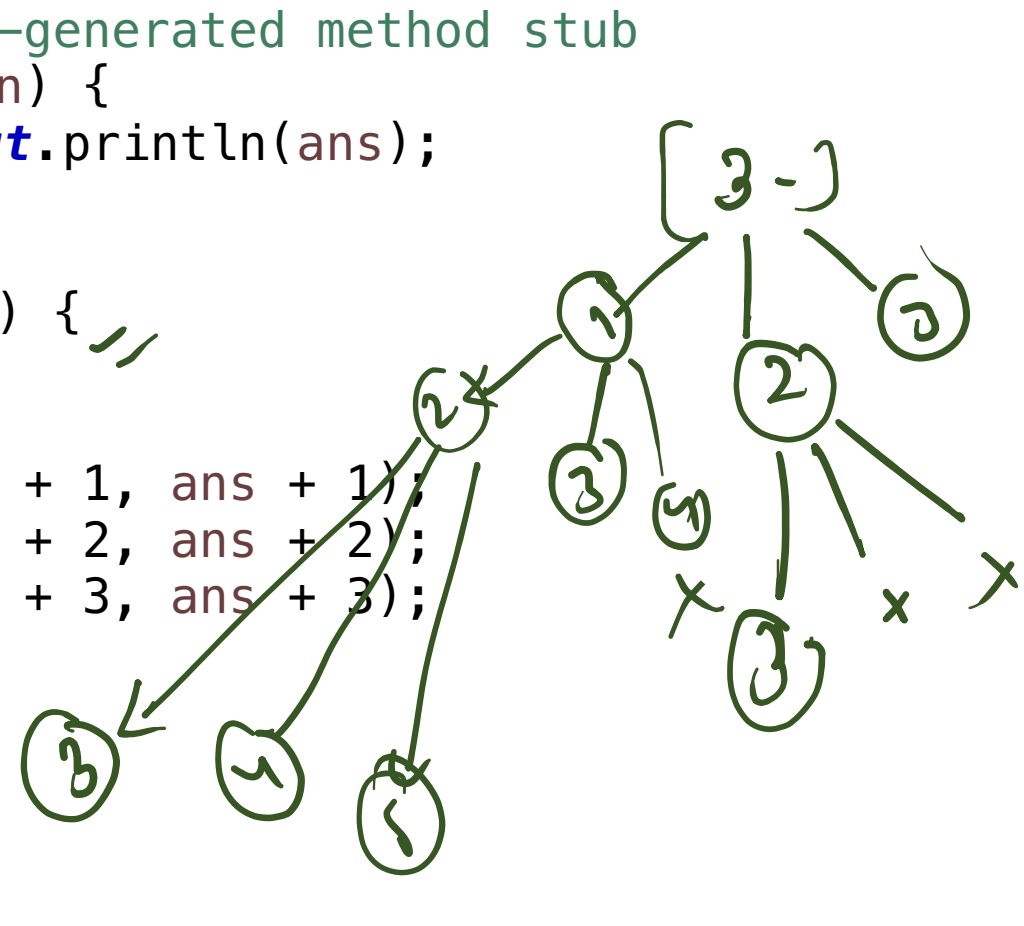
?

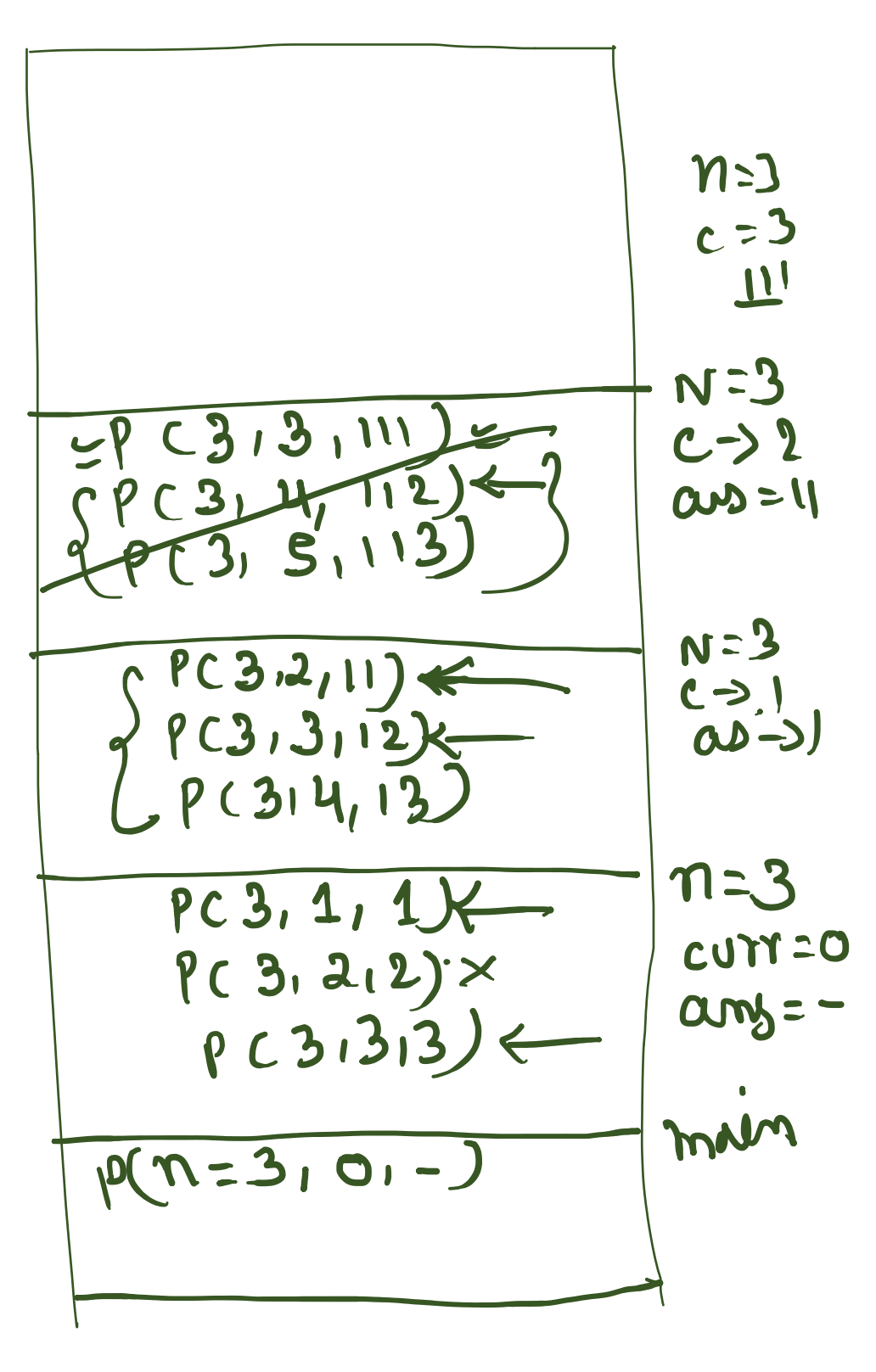
(7, 6, 9)

1111
112
121
211
22
13
31

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 3;
    Path(n, 0, "");
}

public static void Path(int n, int curr, String ans) {
    // TODO Auto-generated method stub
    if (curr == n) {
        System.out.println(ans);
        return;
    }
    if (curr > n) {
        return;
    }
    Path(n, curr + 1, ans + 1);
    Path(n, curr + 2, ans + 2);
    Path(n, curr + 3, ans + 3);
}
```

111
12
21
3

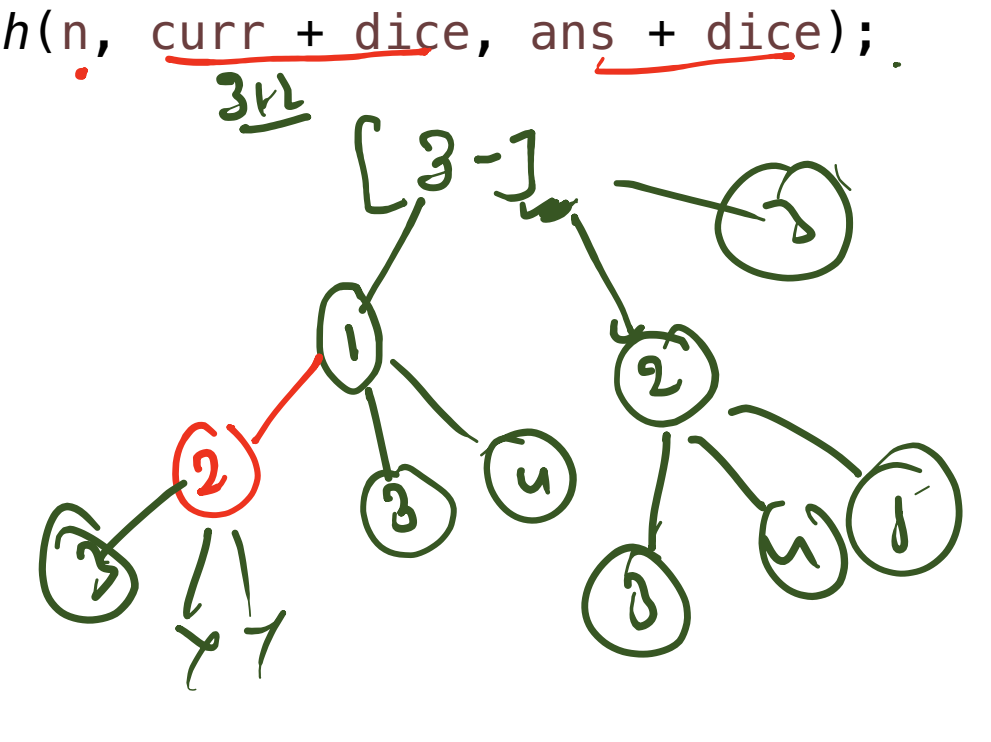

[3 -]
1
2
3
2
3
4
3
4
5
X
X
3

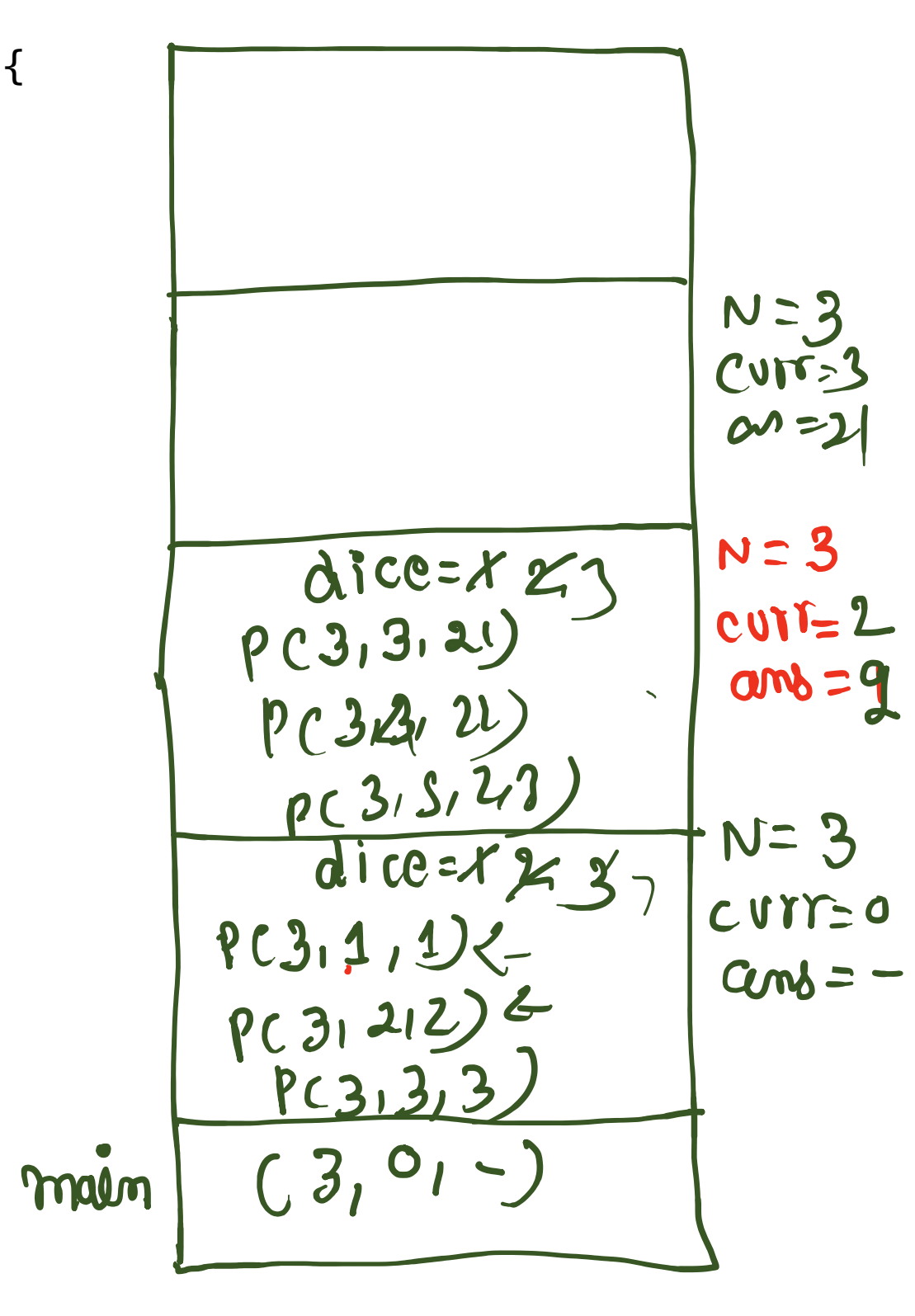

n=3
c=3
111
N=3
c→2
ans=11
P(3, 3, 111)
P(3, 4, 112)
P(3, 5, 113)
N=3
c→1
ans→1
P(3, 2, 11)
P(3, 3, 12)
P(3, 4, 13)
n=3
curr=0
ans=-
P(3, 1, 1)
P(3, 2, 2)
P(3, 3, 3)
main
P(n=3, 0, -)

```
public static void Path(int n, int curr, String ans) {
    // TODO Auto-generated method stub
    if (curr == n) {
        System.out.println(ans);
        return;
    }
    if (curr > n) {
        return;
    }
    for (int dice = 1; dice <=3; dice++) {
        Path(n, curr + dice, ans + dice);
    }
}
```

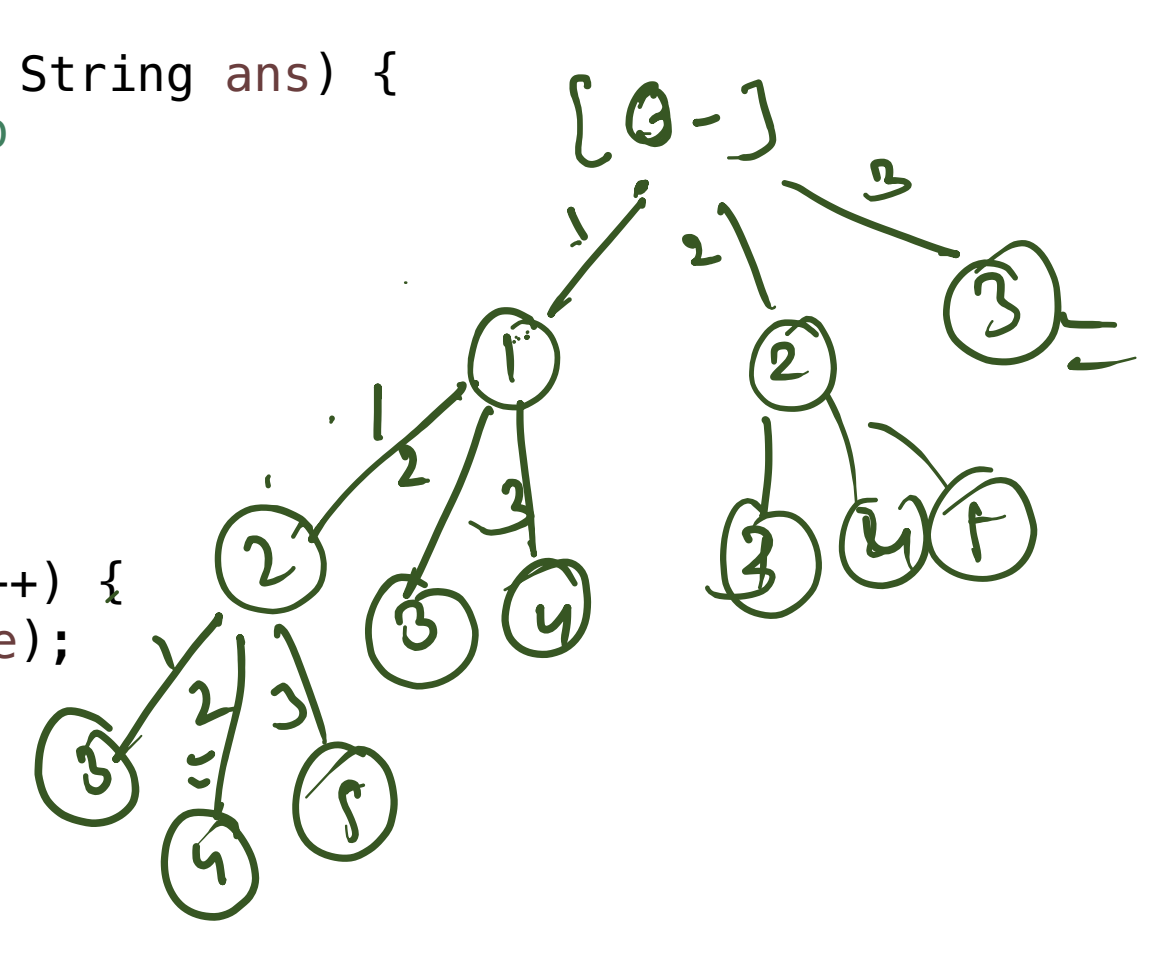
111

3+2


[3 -]
1
2
3
3
4
2
3
4
5
X
Y

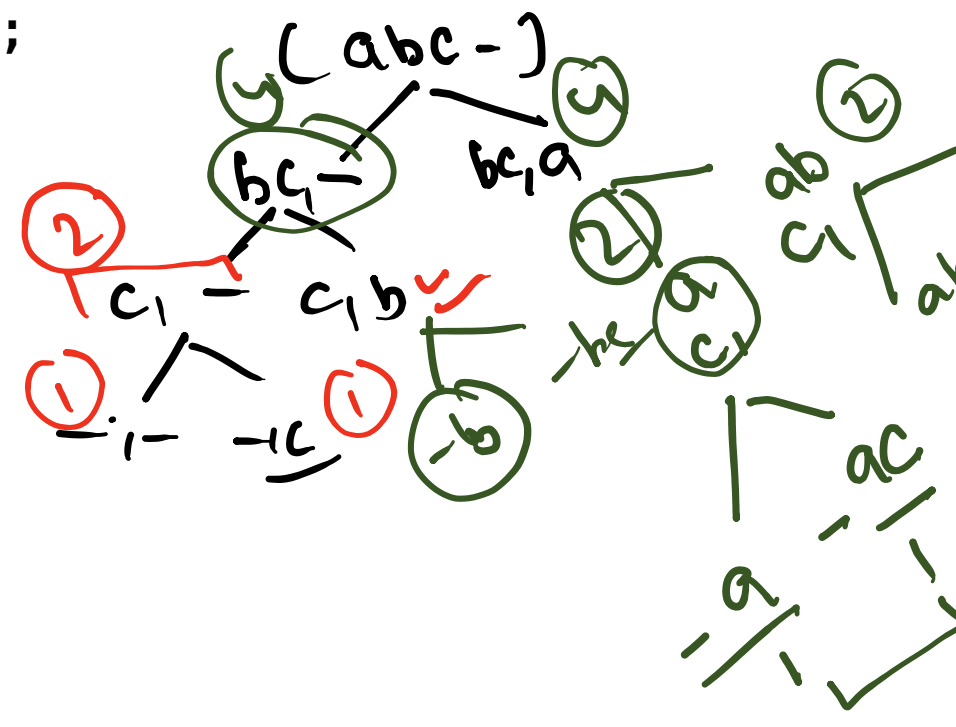

N=3
curr=3
ans=21
N=3
curr=2
ans=2
dice=1 2 3
P(3, 3, 21)
P(3, 4, 22)
P(3, 5, 23)
N=3
curr=0
ans=-
dice=1 2 3
P(3, 1, 1)
P(3, 2, 2)
P(3, 3, 3)
main
(3, 0, -)

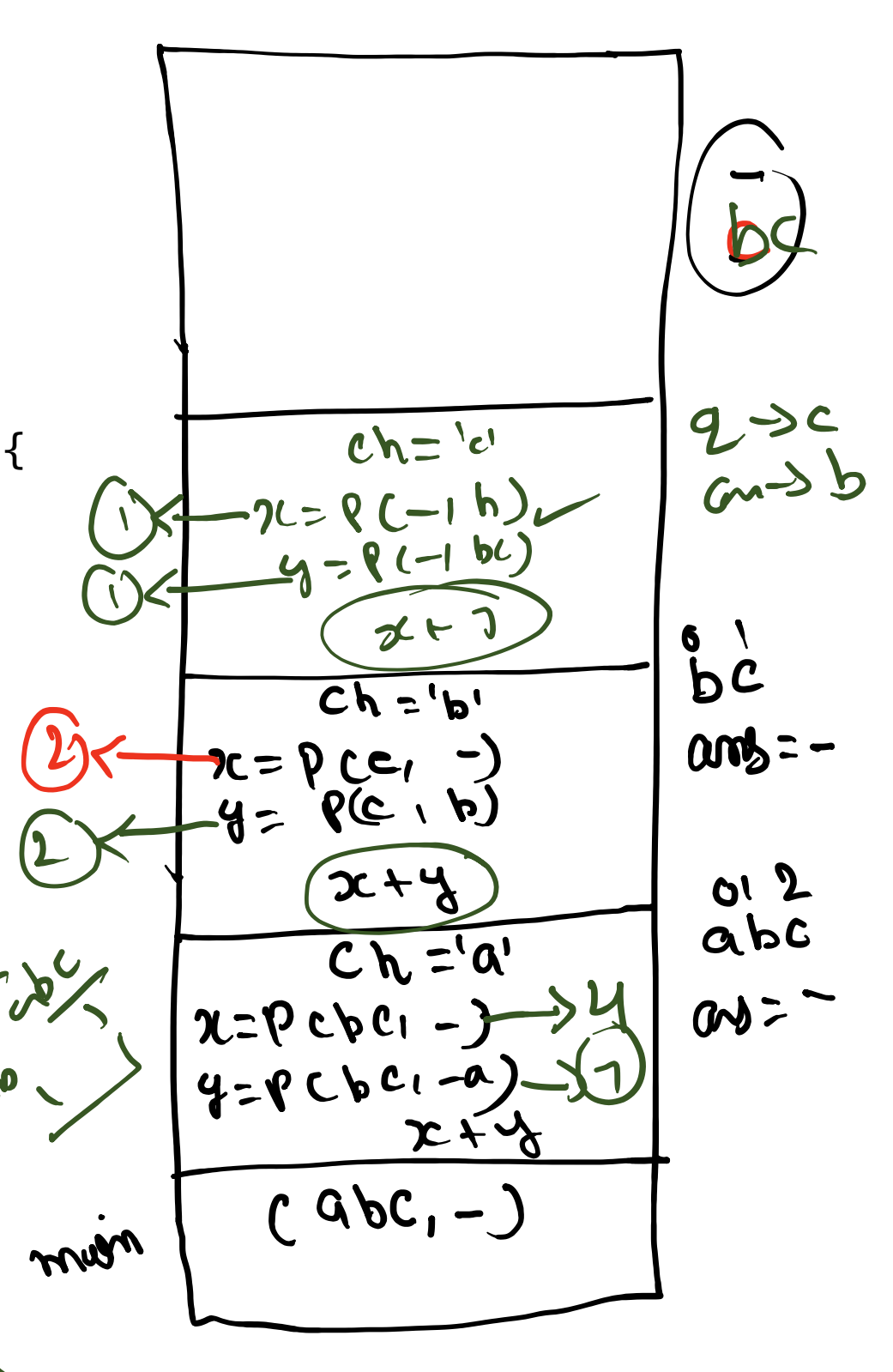
```
public static void Path(int n, int curr, String ans) {
    // TODO Auto-generated method stub
    if (curr == n) {
        System.out.println(ans);
        return;
    }
    if (curr > n) {
        return;
    }
    for (int dice = 1; dice <=3; dice++) {
        Path(n, curr + dice, ans + dice);
    }
}
```


[0 -]
1
2
3
1
2
3
2
3
4
3
4
5
1
2
3
3
4
5

111
12
21

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    String ques = "abc";
    // Print(ques, "");
    System.out.println("\n"+Print(ques, ""));
}

public static int Print(String ques, String ans){
    if (ques.length() == 0) {
        System.out.print(ans + " ");
        return 1;
    }
    char ch = ques.charAt(0);// 'a'
    int x = Print(ques.substring(1), ans);
    int y = Print(ques.substring(1), ans + ch);
    return x + y;
}
```


4
(abc -)
4
bc, -
bc, a
2
2
c, -
c, b
1
1
-, -
-, c
-, b
-, bc
2
ab
c, a
ab
abc
-, a
ac


bc
a→c
ans→b
1
1
ch='c'
x=P(-, b)
y=P(-, bc)
x+y
0 1
bc
ans=-
2
2
ch='b'
x=P(c, -)
y=P(c, b)
x+y
0 1 2
abc
ans=-
ch='a'
x=P(bc, -)→4
y=P(bc, -a)→4
x+y
main
(abc, -)